Our submission combines multitask Chemprop graph neural networks with TabPFN and Kumo models using molecular descriptors, pretrained embeddings and predicted auxiliary assay features.

Models were trained on the public OpenADMET challenge data and curated public CYP datasets. Auxiliary tasks included single-concentration activity, time-dependent inhibition, efficacy and maximum suppression, allowing information from inactive compounds to contribute. Some Chemprop models used an interval-aware loss that penalises predictions only when they fall outside the experimental credible interval.

Model development used five-fold cross-validation and separate holdouts, with evaluation compounds excluded from auxiliary and public training pools. Nonnegative ensembles were constructed independently for each isoform. The final submission combines ensembles selected using two validation surfaces: quota for CYP1A2, CYP2D6 and CYP3A4, and a descriptor-distribution-matched holdout for CYP2C9.

Predictions received conservative affine recalibration to account for population differences between training and blind compounds. CYP3A4 additionally received a regularised monotonic correction fitted using out-of-fold predictions and experimental intervals.

All training data were public, and no blind activity labels were used.